

Muhammad Hamza

AI & Web Developer

Karachi, Pakistan • ihamza.muhammad2003@gmail.com • github.com/MuhammadHamza2210 • linkedin.com/in/muhammad-hamza-962ba1363 • portfolio-lime-rho-96.vercel.app

SUMMARY

Computer Science student (CGPA 3.81/4.0) and self-taught developer who builds AI-powered, full-stack products end to end. Comfortable across the stack — React/TypeScript front ends, Python/Node back ends, and applied AI work like retrieval-augmented generation (RAG) and LLM-integrated features. I turn ideas into shipped, deployed apps and keep learning by building.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL
AI / ML	RAG & LLM app development, Prompt Engineering, LLM APIs (Gemini, Claude, Ollama), CodeBERT, embeddings / vector search
Frontend	React, Vite, Tailwind CSS, Three.js / React Three Fiber, Framer Motion
Backend	Node.js, Express, FastAPI, Flask, REST APIs, WebSockets, JWT auth
Data & Tools	SQLite / SQL, Git & GitHub, Streamlit, Vercel

PROJECTS

Synapse — AI Study OS · React, FastAPI, RAG, Three.js Live: synapse-eight-ruby.vercel.app · github.com/MuhammadHamza2210/Synapse

- Built a retrieval-augmented "Study OS" that ingests a user's own documents into a vector store and answers questions grounded in their material — with citations, not the open web.
- Visualized recalled concepts as an explorable 3D Mind Palace using Three.js for a memorable study experience.

Orion — Real-Time Stock Dashboard · React, FastAPI, WebSockets, AI Live: orion-nine-wine.vercel.app · github.com/MuhammadHamza2210/Orion

- Streamed live prices for 50 tickers over WebSockets with interactive charts and no polling.
- Added on-demand LLM analysis that explains any stock's recent movement in plain language.

CodeScan AI — Code Plagiarism Detector · Python, ML, Streamlit github.com/MuhammadHamza2210/CodeScan-AI

- Combined four engines — AST structural diffing, a trained ML model, winnowing fingerprints and CodeBERT embeddings — for structure-aware similarity scoring that resists variable renaming and reformatting.

LootLo — Full-Stack E-Commerce · Flask, SQLite, REST API github.com/MuhammadHamza2210/Lootlo

- Built a complete commerce flow end to end: user auth, cart, wishlist, coupon codes, live order tracking and an admin console for products and orders.

WildMind AI — AI Wildlife Encyclopedia · React, Vite, Gemini API github.com/MuhammadHamza2210/WildMind-AI

- Conversational encyclopedia returning rich, structured answers for any animal via the Gemini API, with a fast React + Vite interface.

EDUCATION

BS Computer Science — Bahria University, Karachi

2023 – 2027 (expected)

CGPA 3.81 / 4.0 — through 5th semester

Relevant coursework: Data Structures & Algorithms, Databases, Object-Oriented Programming, Software Engineering, Operating Systems.